Problem Set 4

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For every single question, unless otherwise noted I collaborated with Blair Moore, Thomas Li, and John Cho.

Problem 1

1.a

```
df <- read.csv('firms.csv')

y <- df[,'logsalespercap'] |> matrix(byrow = T)
X <- cbind(`(Intercept)` = rep(1, nrow(df)),logemployment = df[,'logemployment'])

solve(t(X)%*%X)%*%t(X)%*%y
```

```
##                         [,1]
## (Intercept)   -6.714174e-07
## logemployment -2.765847e-02
```

1.b

```
df <- read.csv('firms.csv')

y <- df[,'logsalespercap'] |> matrix(byrow = T)

firms <- unique(df$firm)
indicator_mat <- outer(df$firm, firms, function(x,y){x == y})
colnames(indicator_mat) <- paste0('firm_',firms)

X <- cbind(logemployment = df[,'logemployment'], indicator_mat)

(solve(t(X)%*%X)%*%t(X)%*%y)[1,]
```

```
## logemployment
##     0.1174346
```

1.c

Since the fixed effect OLS produced a positive slope, that indicates that within-firm more employment is positively associated with sales. However the negative slope in the pooled regression indicates that the combination of between and within firm variation in employment is negatively related to sales. Noting that the firm-average regression removes all within firm variation, we can loosely conclude that since within is positive, within + between = negative then between = negative. This is a classic simpson's paradox.

Problem 2

2.a

We define:

$$\hat{\beta}_\lambda = \arg\min_{b \in \mathbb{R}} \sum_i (y_i - b)^2 + \lambda b^2$$

Which has b derivative:

$$\sum_i -2(y_i - b) + 2\lambda b$$

Setting equal to zero and solving, we have:

$$\sum_i -2(y_i - b) + 2\lambda b = 0 \implies b = \frac{\sum_i y_i}{(\lambda - n)} \implies b = \frac{n}{\lambda + n}\bar{y}$$

So letting $\eta = 1 - \frac{n}{\lambda+n} = \frac{\lambda}{\lambda+n}$, we can see that the penalized estimator is of the form $(1-\eta)\bar{y}$

2.b

We compute:

$$\begin{aligned}
E[\hat{\beta}] &= E\left[\frac{n}{\lambda+n}\bar{y}\right] \\
&= \frac{n}{\lambda+n} E\left[\bar{y}\right] \\
&= \frac{n}{\lambda+n}\beta \\
&= \frac{n}{\lambda}\eta\beta \\
&= \frac{1-\eta}{\eta}\eta\beta \\
&= (1-\eta)\beta
\end{aligned}$$

Using the fact that n and λ are non-random, and then using the unbiasedness of the sample mean.

The variance can be given by:

$$
\begin{aligned}
E[\hat{\beta}^2] - E[\hat{\beta}]^2 &= E\left[\left(\frac{n}{\lambda+n}\bar{y}\right)^2\right] - \left(\frac{n}{\lambda+n}\right)^2 \beta^2 \\
&= \left(\frac{n}{\lambda+n}\right)^2 E\left[\bar{y}^2\right] - \left(\frac{n}{\lambda+n}\right)^2 \beta^2 \\
&= \left(\frac{n}{\lambda+n}\right)^2 (V(\bar{y}) + E[\bar{y}]^2) - \left(\frac{n}{\lambda+n}\right)^2 \beta^2 \\
&= \left(\frac{n}{\lambda+n}\right)^2 (V(\bar{y}) + \beta^2) - \left(\frac{n}{\lambda+n}\right)^2 \beta^2 \\
&= \left(\frac{n}{\lambda+n}\right)^2 V(\bar{y}) \\
&= \left(\frac{n}{\lambda+n}\right)^2 \frac{\sigma^2}{n} \\
&= (1-\eta)^2 \frac{\sigma^2}{n}
\end{aligned}
$$

Using the standard result on the variance of the sample mean.

2.c

To minimize bias, we'd need $1 - \eta = 1 \implies \eta = 0$. For a fixed n, this corresponds to a minimal payoff of $\lambda = 0$. To minimize variance, we want $\eta = 1$ which corresponds to an infinite penalty.

2.d

An expression for the MSE is:

$$
\begin{aligned}
E\left[\left(\hat{\beta}_\lambda - \beta\right)^2\right] &= E\left[\hat{\beta}_\lambda^2 - 2\hat{\beta}_\lambda\beta + \beta^2\right] \\
&= E\left[\hat{\beta}_\lambda^2\right] - 2\beta E\left[\hat{\beta}_\lambda\right] + \beta^2 \\
&= E\left[\hat{\beta}_\lambda^2\right] - 2(1-\eta)\beta^2 + \beta^2 \\
&= E\left[(1-\eta)^2\bar{y}^2\right] - 2(1-\eta)\beta^2 + \beta^2 \\
&= (1-\eta)^2 E\left[\bar{y}^2\right] - 2(1-\eta)\beta^2 + \beta^2 \\
&= (1-\eta)^2(V(\bar{y}) + E[\bar{y}]^2) - 2(1-\eta)\beta^2 + \beta^2 \\
&= (1-\eta)^2(V(\bar{y}) + \beta^2) - 2(1-\eta)\beta^2 + \beta^2 \\
&= (1-\eta)^2\left(\frac{\sigma^2}{n} + \beta^2\right) - 2(1-\eta)\beta^2 + \beta^2
\end{aligned}
$$

Which has η derivative:

$$
-2(1-\eta)\left(\frac{\sigma^2}{n} + \beta^2\right) + 2\beta^2
$$

Solving the FOC yields:

$$
\boxed{\eta^* = \frac{\sigma^2}{\sigma^2 + n\beta^2}}
$$

But that expression is somewhat unsatisfying since η itself is defined in terms of n solving for the implied λ^*, we get the even nicer $\lambda^* = \sigma^2/\beta^2$.

2.e

Since the family of normal distributions is closed under affine transformations, $y \sim \mathcal{N}(\beta, \sigma^2)$ is the same distribution as $b + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. So express each of our y_is in the form, we have that $y_i = \beta + \varepsilon_i$. The in-sample loss is then:

$$\begin{aligned}
\frac{1}{n}\sum_i (\beta + \varepsilon_i - \hat{\beta}_\lambda)^2 &= \frac{1}{n}\sum_i (\beta + \varepsilon_i - (1-\eta)\overline{y})^2 \\
&= \frac{1}{n}\sum_i (\beta + \varepsilon_i - (1-\eta)\frac{1}{n}\sum_j (\beta + \varepsilon_j))^2 \\
&= \frac{1}{n}\sum_i (\beta + \varepsilon_i - (1-\eta)\beta - (1-\eta)\frac{1}{n}\sum_j (\varepsilon_j))^2 \\
&= \frac{1}{n}\sum_i (\eta\beta + \varepsilon_i - (1-\eta)\frac{1}{n}\sum_j \varepsilon_j)^2 \\
&\equiv \frac{1}{n}\sum_i r_i^2
\end{aligned}$$

Where that last step is just defining r_i to be everything inside the the summation.

The expected loss is then $E\left[\frac{1}{n}\sum_i r_i^2\right]$. Which by linearity expectations is the same $\frac{1}{n}\sum_i E\left[r_i^2\right]$, But since every r_i is constructed with the same distributions, each of their expectations will be the same. So the expected loss reduces to:$\frac{1}{n}nE\left[r_1^2\right] = E\left[r_1^2\right]$. That is to say that the expected loss over the whole sample is the same as the expected loss on an arbitrarily selected unit.

We can then deduce:

$$
\begin{aligned}
E\left[r_1^2\right] &= V(r^2) + E[r]^2 \\
&= V(r^2) + E\left[\eta\beta + \varepsilon_i - (1-\eta)\frac{1}{n}\sum_j \varepsilon_j\right]^2 \\
&= V(r^2) + \left(E\left[\eta\beta\right] + E\left[\varepsilon_i\right] - (1-\eta)E\left[\frac{1}{n}\sum_j \varepsilon_j\right]\right)^2 \\
&= V(r^2) + \left(\eta\beta + 0 - (1-\eta)0\right)^2 \\
&= V(r^2) + (\eta\beta)^2 \\
&= V\left(\eta\beta + \varepsilon_i - (1-\eta)\frac{1}{n}\sum_j \varepsilon_j\right) + (\eta\beta)^2 \\
&= V\left(\varepsilon_i - (1-\eta)\frac{1}{n}\sum_j \varepsilon_j\right) + (\eta\beta)^2 \\
&= V\left(\varepsilon_i - \frac{(1-\eta)}{n}\varepsilon_i - \frac{(1-\eta)}{n}\sum_{j\neq i} \varepsilon_j\right) + (\eta\beta)^2 \\
&= V\left(\frac{n-(1-\eta)}{n}\varepsilon_i - \frac{(1-\eta)}{n}\sum_{j\neq i} \varepsilon_j\right) + (\eta\beta)^2 \\
&= V\left(\frac{n-(1-\eta)}{n}\varepsilon_i\right) + V\left(\frac{(1-\eta)}{n}\sum_{j\neq i} \varepsilon_j\right) + (\eta\beta)^2 && \text{(made to be indep.)} \\
&= \left(\frac{n-(1-\eta)}{n}\right)^2\sigma^2 + V\left(\frac{(1-\eta)}{n}\sum_{j\neq i} \varepsilon_j\right) + (\eta\beta)^2 \\
&= \left(\frac{n-(1-\eta)}{n}\right)^2\sigma^2 + \left(\frac{(1-\eta)}{n}\right)^2(n-1)\sigma^2 + (\eta\beta)^2 \\
&= \frac{\sigma^2}{n^2}\left[\left(n-(1-\eta)\right)^2 + \left((1-\eta)\right)^2(n-1)\right] + (\eta\beta)^2 \\
&= \frac{\sigma^2}{n^2}\left[\left((n-1)+\eta\right)\right)^2 + \left((1-\eta)\right)^2(n-1)\right] + (\eta\beta)^2 \\
&= \frac{\sigma^2}{n^2}\left[(n-1)^2 + 2\eta(n-1) + \eta^2 + (1-2\eta-\eta^2)(n-1)\right] + (\eta\beta)^2 \\
&= \frac{\sigma^2}{n^2}\left[(n-1)^2 + 2\eta(n-1) + \eta^2 + (n-1) - 2\eta(n-1) - \eta^2(n-1)\right] + (\eta\beta)^2 \\
&= \frac{\sigma^2}{n^2}\left[(n-1)^2 + \eta^2 + (n-1) - \eta^2(n-1)\right] + (\eta\beta)^2 \\
&= \frac{\sigma^2}{n^2}\left[(n-1)((n-1)+1) + \eta^2 - \eta^2(n-1)\right] + (\eta\beta)^2 \\
&= \frac{\sigma^2}{n^2}\left[(n-1)n + \eta^2 - \eta^2(n-1)\right] + (\eta\beta)^2 \\
&= \frac{\sigma^2}{n^2}\left[(n-1)n + n\eta^2\right] + (\eta\beta)^2 \\
&= \frac{\sigma^2}{n}\left[(n-1) + \eta^2\right] + (\eta\beta)^2
\end{aligned}
$$

For the out of sample error, assuming correct specification things are even easier. Representing y_{n+1} as $y_{n+1} = \beta + \varepsilon_{n+1}$, then we can express

$$
\begin{aligned}
E[(y_{n+1} - \hat{\beta}_\lambda)^2] &= E[(y_{n+1} - (1-\eta)\overline{y})^2] \\
&= E[y_{n+1}^2 - 2(1-\eta)y_{n+1}\overline{y} + (1-\eta)^2\bar{y}^2] \\
&= E[y_{n+1}^2] - 2(1-\eta)E[y_{n+1}\overline{y}] + (1-\eta)^2 E[\bar{y}^2] \\
&= E[y_{n+1}^2] - 2(1-\eta)E[y_{n+1}]E[\overline{y}] + (1-\eta)^2 E[\bar{y}^2], \text{bc indep} \\
&= E[y_{n+1}^2] - 2(1-\eta)\beta^2 + (1-\eta)^2 E[\bar{y}^2] \\
&= V(y_{n+1}) + E[y_{n+1}]^2 - 2(1-\eta)\beta^2 + (1-\eta)^2 E[\bar{y}^2] \\
&= \sigma^2 + \beta^2 - 2(1-\eta)\beta^2 + (1-\eta)^2 E[\bar{y}^2] \\
&= \sigma^2 + (1 - 2(1-\eta))\beta^2 + (1-\eta)^2 E[\bar{y}^2] \\
&= \sigma^2 + (-1+2\eta)\beta^2 + (1-\eta)^2 E[\bar{y}^2] \\
&= \sigma^2 + (-1+2\eta)\beta^2 + (1-\eta)^2 V[\bar{y}] + (1-\eta)^2 E[\bar{y}]^2 \\
&= \sigma^2 + (-1+2\eta)\beta^2 + (1-\eta)^2 \frac{\sigma^2}{n} + (1-\eta)^2\beta^2 \\
&= \sigma^2 + (-1+2\eta+(1-\eta)^2)\beta^2 + (1-\eta)^2\frac{\sigma^2}{n} \\
&= \sigma^2 + (-1+2\eta+1-2\eta+\eta^2)\beta^2 + (1-\eta)^2\frac{\sigma^2}{n} \\
&= \sigma^2 + \eta^2\beta^2 + (1-\eta)^2\frac{\sigma^2}{n} \\
&= (n + (1-\eta)^2)\frac{\sigma^2}{n} + \eta^2\beta^2 \\
&= \left(1 + \frac{(1-\eta)^2}{n}\right)\sigma^2 + \eta^2\beta^2
\end{aligned}
$$

And with some (ommited) algebra, one can show that in sample loss is smaller than out of sample loss whenever $\eta \leq 1$. And since $\eta = \frac{\lambda}{n+\lambda}$ and $n \geq 0, \lambda \geq 0$, this condition will always hold. We always perform better in sample than out. This kinda makes sense.